CTU-Chat: A Hybrid Retrieval-Augmented Generation System for University Student Services

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Abstract. University administrative question answering demands more than semantic similarity matching: answers must respect document validity, organizational authority, temporal scope, and exact procedural or numerical rules. This paper presents CTU-Chat, a bilingual Vietnamese–English framework that integrates metadata-filtered BM25 and dense retrieval, a provenance-preserving knowledge graph, deterministic evidence reranking, and a specialist-agent workflow. PostgreSQL serves as the authoritative data store, while Qdrant and Neo4j function as rebuildable retrieval projections; every candidate is revalidated against publication, approval, audience, category, and effective-date constraints prior to generation. The framework is evaluated through five cumulative variants—from hybrid retrieval (M0) to coverage-aware agentic GraphRAG (M4)—using `TBD` human-verified questions spanning `TBD` administrative categories. Ablations quantify the marginal contributions of graph retrieval, evidence reranking, quality scoring, and coverage-aware retry. This work provides a reproducible design for evidence-grounded administrative assistants in low-resource bilingual settings.

Keywords: Retrieval-augmented generation · GraphRAG · Agentic AI · Evidence provenance · Hybrid retrieval · Bilingual question answering · University administration · Vietnamese NLP

1 Introduction

Providing accurate and timely administrative support is an essential component of modern university student services. Students frequently seek information about tuition fees, scholarships, student loans, academic regulations, curricula, and administrative procedures. However, these answers are distributed across versioned regulations, official announcements, fee tables, and curriculum documents, making reliable information retrieval a challenging task. Incorrect or outdated responses may directly affect students' academic and financial decisions.

Recent advances in Retrieval-Augmented Generation (RAG) [4] have significantly improved question answering by grounding large language models on external knowledge. Dense retrieval effectively captures semantic similarity, while lexical retrieval remains valuable for matching exact identifiers such as course codes, decision numbers, and Vietnamese administrative terminology. Knowledge graphs and agent-based orchestration further extend RAG by supporting relational reasoning and specialized tool invocation.

Despite these advances, existing RAG systems remain insufficient for university administrative services. First, semantic retrieval alone often fails to retrieve exact policy identifiers and structured tabular information. Second, most systems do not preserve document governance, including version validity, publication status, and effective dates. Third, they lack deterministic mechanisms for numerical calculations such as tuition reduction and scholarship eligibility, leading to fluent but potentially incorrect responses. These limitations reveal a research gap in building evidence-grounded and governance-aware RAG systems for higher education.

Research Question. *How can a hybrid retrieval framework combining lexical search, dense retrieval, knowledge graphs, and deterministic tools improve the reliability and factual accuracy of university student-service question answering?*

To address this question, this paper proposes **CTU-Chat**, a hybrid Retrieval-Augmented Generation framework for Can Tho University. The proposed system integrates Vietnamese BM25, dense vector retrieval, Reciprocal Rank Fusion (RRF), metadata-governed evidence filtering, graph-based relational retrieval, and intent-aware multi-agent orchestration with typed calculation tools. This architecture preserves evidence provenance while providing accurate, transparent, and reproducible answers for administrative and academic queries.

The main contributions of this work are as follows:

- A governance-aware hybrid retrieval architecture that combines lexical, semantic, and graph-based evidence under temporal and metadata constraints.
- An intent-aware multi-agent framework integrating Neo4j graph retrieval with deterministic academic and financial calculation tools.
- A comprehensive experimental evaluation on a human-verified university student-service benchmark to assess retrieval quality, answer correctness, and system reliability.

2 Related Work

This section reviews representative studies on Retrieval-Augmented Generation, hybrid retrieval, GraphRAG, agentic AI, and educational question answering. Rather than summarizing previous work, we analyze their limitations and identify the research gap addressed by CTU-Chat.

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG), first introduced by Lewis et al. [4], combines external document retrieval with large language models to generate evidence-grounded responses. By conditioning generation on retrieved passages, RAG significantly improves factual accuracy and reduces hallucinations in open-domain question answering.

However, conventional RAG assumes a static and homogeneous document collection in which all retrieved documents are treated as equally reliable. Such an assumption is unsuitable for university administrative services because regulations are continuously revised and possess different publication states and periods of validity. Consequently, existing RAG systems rarely preserve document provenance or distinguish obsolete documents from currently effective regulations, limiting their reliability in governance-oriented applications.

2.2 Lexical and Dense Retrieval for Bilingual Text

Lexical retrieval and dense retrieval represent two complementary paradigms for information retrieval. BM25 [6] remains highly effective for matching exact policy terminology, course identifiers, and administrative codes, while multilingual dense embeddings [1] provide robust semantic matching across Vietnamese and English paraphrases.

The BM25 ranking function is defined as:

$$s_{\mathrm{BM25}}(q,d) = \sum_{w \in q} \mathrm{IDF}(w) \frac{\mathrm{tf}(w,d)(k_1+1)}{\mathrm{tf}(w,d) + k_1\left(1 - b + b\,\frac{|d|}{\overline{|d|}}\right)}. \tag{1}$$

Dense retrieval computes semantic similarity using cosine similarity between query and document embeddings:

$$s_{\mathrm{dense}}(q,d) = \frac{\mathbf{h}_q^{\top}\mathbf{h}_d}{\|\mathbf{h}_q\|_2 \|\mathbf{h}_d\|_2}. \tag{2}$$

To leverage the strengths of both paradigms, hybrid retrieval commonly employs Reciprocal Rank Fusion (RRF) [2] to merge lexical and semantic rankings without score normalization.

Despite their complementary advantages, existing hybrid retrieval methods primarily optimize relevance while overlooking governance metadata. Highly ranked passages may therefore originate from expired or unauthorized documents, and Vietnamese-specific challenges such as word segmentation and compound administrative terminology remain insufficiently addressed by general multilingual retrieval models.

2.3 Knowledge Graphs and GraphRAG

Knowledge graphs extend retrieval by explicitly modeling entities and their relationships, enabling multi-hop reasoning across organizational structures, curricula, prerequisites, and administrative procedures. GraphRAG [3] integrates graph traversal with retrieval-augmented generation to discover evidence beyond isolated text chunks.

Nevertheless, most GraphRAG frameworks directly project LLM-extracted facts into the graph without validating their correctness. Unreviewed graph edges may propagate inaccurate relationships during retrieval, while graph-expanded candidates frequently bypass the governance constraints enforced in the authoritative document repository. These issues reduce the trustworthiness of graph reasoning in highly regulated domains such as university administration.

2.4 Agentic RAG and Multi-Agent Systems

Agentic RAG extends traditional retrieval pipelines through autonomous planning, tool invocation, and iterative evidence acquisition [7]. Supervisor–Worker orchestration frameworks, such as LangGraph, further improve scalability by assigning specialized responsibilities to domain-specific agents with dedicated tools and contextual memory.

Although these architectures provide greater flexibility than conventional RAG, increased autonomy also introduces new sources of uncertainty. Routing errors, inappropriate tool selection, memory leakage across sessions, and non-deterministic reasoning negatively affect reproducibility. Moreover, existing studies rarely evaluate whether an agent should abstain or retry retrieval when supporting evidence is insufficient, leaving reliability largely unexplored.

2.5 Administrative and Educational Question Answering

Educational question answering systems have been widely applied to university student services, including academic advising, curriculum consultation, and administrative assistance [5]. These systems demonstrate the practicality of conversational AI for accessing institutional knowledge and supporting student services.

However, most existing educational QA systems remain document-centric rather than governance-centric. They seldom support bilingual Vietnamese–English administrative language, preserve citation provenance, or integrate deterministic calculation tools for tuition and GPA policies. In addition, multi-agent routing and evidence-aware response generation are rarely evaluated, indicating that reliable university governance remains an open research problem.

2.6 Research Gap and Problem Formulation

The literature review reveals that no existing approach simultaneously addresses multilingual retrieval, document governance, provenance preservation, temporal validity, and deterministic policy computation within a unified university student-service framework. This study therefore formulates the administrative QA problem as follows.

Let a user query be defined as $q = (x, \ell, \mathcal{C}, \mathcal{A}, t)$, where x is the question text, $\ell \in \{vi, en\}$ is the language, $\mathcal{C}$ denotes category constraints, $\mathcal{A}$ represents an optional audience constraint, and t is the reference time for answer validity.

The objective is to generate an answer y together with a set of supporting citations $Z \subseteq E$ such that every cited evidence is relevant, originates from an approved and effective document version, satisfies governance constraints, and fully supports the generated claims. When sufficient evidence is unavailable, the system should abstain or explicitly express uncertainty rather than generating unsupported information.

The governance constraint is formalized by the predicate:

$$G(e,q) = I_{\text{approved}}(e) \cdot I_{\text{published}}(e) \cdot I_{\text{category}}(e,\mathcal{C}) \cdot I_{\text{audience}}(e,\mathcal{A}) \cdot I_{\text{effective}}(e,t). \tag{3}$$

Only evidence satisfying $G(e,q) = 1$ is allowed to participate in answer generation. This formulation establishes the theoretical foundation for the governance-aware hybrid retrieval architecture proposed in the following section.

3 Proposed System Architecture

4 Methodology

This section describes the core methodologies implemented in CTU-Chat, including hybrid retrieval, deterministic evidence reranking, evidence-quality assessment, coverage-aware retrieval, and domain-specific deterministic calculation tools.

4.1 Hybrid Retrieval with Reciprocal Rank Fusion

To improve retrieval robustness, CTU-Chat executes three complementary retrieval strategies concurrently: BM25 lexical retrieval, dense semantic retrieval, and graph-based retrieval. Since these retrievers produce heterogeneous score distributions, their scores are not directly summed. Instead, retrieved candidates are merged, deduplicated by `chunk_id`, and ranked using Reciprocal Rank Fusion (RRF).

For the retrieval source set $\mathcal{S} = \{\text{BM25}, \text{Dense}, \text{Graph}\}$, the RRF score of a candidate document d is computed as

$$s_{\text{RRF}}(d) = \sum_{s \in \mathcal{S}_d} \frac{1}{k + r_s(d)}, \qquad k = 60, \tag{4}$$

where $r_s(d)$ denotes the ranking position of document d returned by retrieval source s, and $\mathcal{S}_d$ contains only the retrievers that successfully retrieve d. The constant $k = 60$ is fixed across all experiments to ensure reproducibility.

For relational questions, graph retrieval expands evidence from a Neo4j full-text seed. Let S_q denote the initial seed set for query q. The expanded neighborhood is defined as

$$\mathcal{N}(S_q) = \mathcal{N}_{\text{next}}(S_q) \cup \mathcal{N}_{\text{org}}(S_q) \cup \mathcal{N}_{\text{entity}}(S_q), \tag{5}$$

where $\mathcal{N}_{\text{next}}$, $\mathcal{N}_{\text{org}}$, and $\mathcal{N}_{\text{entity}}$ represent adjacent chunks, co-issuing organizations, and shared validated entities, respectively. The graph is treated as an evidence expansion mechanism rather than an authority source, and every candidate must satisfy the governance predicate $G(e, q)$ before answer generation.

4.2 Deterministic Evidence Reranking

Following RRF fusion, CTU-Chat applies a deterministic reranking strategy to improve citation stability and ranking reproducibility. Unlike neural rerankers, the proposed method relies on interpretable lexical and structural features, ensuring that identical inputs always produce identical rankings.

The reranking score is defined as

$$s_R(d, q) = 0.55\, R_c + 0.25\, R_h + 0.10\, I_{\text{all}} + 0.10\, C_s, \tag{6}$$

where R_c denotes query-term recall within the document content, R_h measures recall in the heading hierarchy, I_{all} indicates complete keyword coverage, and $C_s = |\mathcal{S}_d|/3$ represents cross-source consensus. Documents with identical scores are further ordered deterministically using RRF score, source agreement, heading overlap, document identifier, and chunk UUID.

4.3 Evidence-Quality Assessment

Beyond relevance ranking, CTU-Chat estimates the overall reliability of each evidence item before generation. The proposed evidence-quality score jointly considers semantic relevance, organizational authority, temporal freshness, provenance completeness, and consistency across multiple retrieval sources.

$$Q(e) = 0.35\, R_e + 0.20\, A_e + 0.15\, F_e + 0.20\, P_e + 0.10\, C_e, \tag{7}$$

where R_e, A_e, F_e, P_e, and C_e denote relevance, authority, freshness, provenance, and consistency, respectively. All component scores are retained for subsequent analysis and are later utilized by the coverage-aware retrieval mechanism.

4.4 Coverage-Aware Retrieval

A reliable administrative question answering system should avoid generating responses when supporting evidence is incomplete. Therefore, CTU-Chat evaluates how comprehensively the retrieved evidence satisfies the facts required by a user query.

Let F_q denote the set of required facts inferred from query q, and let $F_E \subseteq F_q$ be the subset supported by the current evidence set E. The coverage score is computed as

$$\text{Coverage}(q, E) = \begin{cases} 1, & |F_q| = 0, \\ \dfrac{|F_E|}{|F_q|}, & \text{otherwise.} \end{cases} \tag{8}$$

When the coverage score falls below the predefined threshold θ, the framework initiates an additional retrieval round if the retry budget remains available. Otherwise, the system abstains from unsupported answer generation and explicitly reports insufficient evidence.

4.5 Deterministic Academic and Financial Tools

Certain university services require exact numerical computation rather than language-model reasoning. CTU-Chat therefore integrates typed deterministic tools for course grading, GPA calculation, and tuition fee estimation, ensuring reproducible results that are directly traceable to institutional policies.

Course Grade Calculation Given assessment scores x_i and weights w_i satisfying $\sum_i w_i = 100$, the final 10-point score is calculated as

$$S_{10} = \mathrm{Round}_{0.1}\left(\sum_i \mathrm{Round}_{0.1}(x_i)\frac{w_i}{100}\right). \tag{9}$$

The resulting score is converted into the official CTU letter-grade scale (A, B+, B, C+, C, D+, D, and F).

GPA Calculation For eligible courses $\mathcal{I}$ with credits c_i and grade points g_i, GPA is computed as

$$\mathrm{GPA} = \mathrm{Round}_{0.01}\left(\frac{\sum_{i\in\mathcal{I}} c_i g_i}{\sum_{i\in\mathcal{I}} c_i}\right). \tag{10}$$

Courses with statuses Missing (M), Incomplete (I), and Withdrawn (W) are excluded according to the university grading policy.

Tuition Fee Calculation For each registered course i, the payable tuition fee is determined from the credit load, tuition rate, exemption basis, and reduction percentage:

$$\mathrm{Gross}_i = c_i r_i m_i, \tag{11}$$

$$\mathrm{Reduction}_i = c_i b_i \frac{p_i}{100}, \tag{12}$$

$$\mathrm{Payable}_i = \mathrm{Round}_{1\,\mathrm{VND}}\left(\max(0, \mathrm{Gross}_i - \mathrm{Reduction}_i)\right). \tag{13}$$

All computations use decimal arithmetic, and every tool output preserves the corresponding evidence identifier and policy reference to guarantee auditability.

5 Experimental Results

This section presents the experimental dataset, implementation environment, evaluation metrics, experimental scenarios, and discussion of the proposed CTU-Chat framework.

5.1 Experimental Dataset

A benchmark dataset was constructed to evaluate the proposed CTU-Chat framework on university student-service question answering. The dataset was developed from official Can Tho University documents, including academic regulations, training programs, tuition policies, scholarships, student loan policies, social support, dormitory regulations, and administrative procedures.

Instead of manually composing every question, candidate questions were first generated using Claude based on the collected documents. The generated dataset then underwent a two-stage validation process. First, a validation notebook was used to verify the availability of supporting documents and evidence references. Second, all questions were manually reviewed to remove duplicated or ambiguous samples and to ensure that each question had a complete ground-truth answer. A final validation step was performed before constructing the benchmark.

The final benchmark contains 100 representative questions selected according to three criteria: (1) coverage of all major administrative categories, (2) high-frequency real-world student inquiries, and (3) complete ground-truth evidence for every sample.

Table 1. Distribution of the benchmark dataset.

Category	Questions
Actual Tuition	14
Scholarship	14
Academic Rules	14
Student Loan	12
Social Support	11
General Regulations	9
Academic Program	9
Exemption Policy	9
Exemption Basis	5
Other	3
Total	**100**

5.2 Experimental Tools

The proposed system was implemented using a hybrid retrieval architecture consisting of vector, graph, and relational databases. Dense embeddings were generated using BGE-M3, while LangGraph was employed for multi-agent orchestration. System performance was evaluated using the RAGAS framework.

Table 2. Experimental implementation environment.

Component	Technology
Programming language	Python 3.12
Vector database	Qdrant
Graph database	Neo4j
Metadata database	PostgreSQL
Embedding model	BGE-M3
LLM	Qwen3-Next-80B
Agent framework	LangGraph
Evaluation framework	RAGAS

5.3 Evaluation Metrics

Two groups of evaluation metrics were adopted in this study.

Retrieval metrics evaluate the quality of evidence retrieval, including Hit@1 (H@1), Hit@3 (H@3), Precision@5 (P@5), Recall@5, and Mean Reciprocal Rank (MRR@10).

End-to-end QA metrics were evaluated using the RAGAS benchmark, including Answer Relevancy (AR), Context Recall (CR), Context Precision (CP), and Answer Correctness (AC). All metrics range from 0 to 1, where higher values indicate better performance.

5.4 Experimental Scenarios

Two groups of experiments were conducted using the same 100-question benchmark. The first evaluates the retrieval module (E1–E5), while the second evaluates the complete question answering pipeline (T1–T7).

Scenario A: Retrieval Performance Five cumulative retrieval configurations (E1–E5) were designed to evaluate the contribution of dense retrieval, hybrid retrieval, graph expansion, deterministic reranking, and governance-aware filtering.

The retrieval experiment investigates whether each additional component improves the quality of evidence retrieval. As shown in Table 3, E4 achieved the highest Recall@5 (0.889) and MRR@10 (0.816), demonstrating that hybrid retrieval combined with graph expansion provides the most effective retrieval strategy. Although E5 preserves the same Recall@5, its ranking quality slightly decreases, indicating that governance-aware filtering introduces a small trade-off in retrieval ranking.

Table 3. Retrieval evaluation results of E1–E5.

Config.	H@1	H@3	P@5	R@5	MRR
E1	0.444	0.667	0.156	0.778	0.573
E2	0.222	0.222	0.044	0.222	0.238
E3	0.222	0.556	0.133	0.667	0.435
E4	0.778	0.778	0.178	0.889	0.816
E5	0.667	0.778	0.178	0.889	0.750

Scenario B: End-to-End Question Answering The second experiment evaluates seven complete QA configurations (T1–T7) using the RAGAS benchmark. This scenario measures not only retrieval quality but also the factual correctness and relevance of generated answers under different retrieval and generation settings.

Table 4 shows that T4 achieved the strongest overall performance, obtaining the highest Context Recall (0.922) and Context Precision (0.817), while maintaining a competitive Answer Correctness score of 0.751. In contrast, T7 produced the highest Answer Relevancy (0.813) but suffered a substantial decrease in Answer Correctness (0.459), suggesting that semantically relevant responses do not necessarily guarantee factually correct answers.

Table 4. RAGAS evaluation results of T1–T7.

Config.	AR	CR	CP	AC
T1	0.532	0.618	0.516	0.598
T2	0.735	0.850	0.742	0.739
T3	0.739	0.860	0.657	0.725
T4	0.811	0.922	0.817	0.751
T5	0.675	0.790	0.604	0.675
T6	0.810	0.912	0.815	0.752
T7	0.813	0.912	0.815	0.459

5.5 Discussion

The experimental results demonstrate that retrieval quality has a direct impact on the overall performance of the QA system. Compared with the baseline, combining lexical, dense, and graph retrieval significantly improves evidence coverage, leading to higher Recall@5 and MRR in the retrieval experiments and higher Context Recall and Context Precision in the RAGAS evaluation.

The retrieval experiments identify E4 as the most effective configuration, whereas the end-to-end evaluation shows that T4 consistently achieves the best overall balance between retrieval quality and answer generation. These findings support the effectiveness of the proposed governed hybrid retrieval framework with deterministic reranking.

Several limitations should be acknowledged. First, the benchmark questions were generated with the assistance of Claude and subsequently verified through human review, which may not fully represent real student query distributions. Second, RAGAS relies on LLM-based evaluation and may introduce evaluator variance. Finally, the experiments were conducted using documents from a single institution, and future work should validate the framework on multi-university administrative datasets.

6 Conclusion

This paper presented **CTU-Chat**, a hybrid Retrieval-Augmented Generation (RAG) system for university student services that integrates lexical, dense, and graph retrieval with governed document management for bilingual Vietnamese–English administrative question answering.

The proposed framework addresses five research gaps identified in existing educational RAG systems, namely the absence of document governance, limited integration of lexical, semantic, and graph retrieval, insufficient provenance preservation, lack of evidence-aware retrieval, and the reliance on LLM-based numerical reasoning for academic and financial services.

Experimental results demonstrate that the proposed system effectively addresses these limitations. In the retrieval evaluation, the E4 configuration achieved the best overall performance with a Recall@5 of 0.889 and an MRR@10 of 0.816, confirming that hybrid retrieval and graph expansion significantly improve evidence coverage and retrieval quality. In the end-to-end evaluation, T4 obtained the highest Context Recall (0.922), Context Precision (0.817), and an Answer Correctness score of 0.751, indicating that improved retrieval directly contributes to more faithful and accurate answer generation.

Furthermore, the comparison between T4 and T7 shows that high semantic relevance alone does not guarantee factual correctness, highlighting the importance of governed evidence validation and deterministic reasoning for reliable university administrative question answering.

Future work will focus on expanding the benchmark to multiple universities, constructing larger human-authenticated bilingual datasets, evaluating learned cross-encoder rerankers, and enhancing graph construction for more complex multi-hop administrative reasoning.

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